FUNCTIONAL SPECIFICATION DOCUMENT

(FSD)

Automated Network Request Management in ServiceNow

Project Name	Automated Network Request Management in ServiceNow
Prepared By	
Platform	ServiceNow
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1. Introduction

Project Title: Automated Network Request Management in ServiceNow

Project Owner / Contributor:

This Functional Specification Document adapts the supplied full-stack documentation format to the actual ServiceNow implementation. Where the supplied template refers to React, Node.js, Express.js, MongoDB, npm commands or REST endpoints, the equivalent ServiceNow platform component is documented instead; unsupported technologies are not claimed.

2. Project Overview

Purpose

Streamline the end-to-end lifecycle of network-related service requests through a centralized ServiceNow workflow. The solution captures structured requester information, validates and transfers catalog variables, creates an operational record, routes the request for approval, branches according to the approval decision, updates request status, and provides automated notifications and traceability.

Key Features

• Service Catalog-based Network Request submission under Networks and Connectivity.

• Structured catalog variables for Requested For, Mobile Number, Type of Connection, Existing ID, Total Amount, Mode of Payment, Address, Database Number, and Request Number.

• Requester Information variable set and configured dynamic field behaviour using a Catalog UI Policy.

• Flow Designer-based request processing and catalog-variable retrieval.

• REQ/RITM request tracking and lifecycle visibility.

• Configured Database Tables (u_database_tables) operational record for structured network-request data.

• Approval processing through the Ask for Approval action and approval-status branching.

• Automated request-created and approved/resolved email notification actions.

• Roles and ACL-based access control on the configured custom table.

• Approval Request relationship for connecting the operational record with approval records.

• Flow Designer Execution Details / platform logs for troubleshooting.

3. Architecture

Figure 1. ServiceNow solution architecture

Requester / User	**ServiceNow Service Portal**	**Network Request Catalog**	**Catalog Variables / UI Policy**	**Flow Designer**
Approval	**REQ / RITM**	**Database Tables (u_database_tables)**	**Notifications**	**Operational / Fulfilment**
Roles & ACLs	**Approval Request Relationship**	**Execution Details / Logs**	**Current Scope**	**Future Extensions**

Frontend / User Interface

The user-facing layer is the ServiceNow Service Portal and Service Catalog. The requester opens the Network Request catalog item, enters the configured request variables, and submits the request. UI Policies provide the configured dynamic field behaviour.

Backend / Application Logic

The application-logic equivalent is ServiceNow platform configuration. Flow Designer is the primary automation engine. It retrieves catalog variables, creates the configured Database Tables record, initiates approval, evaluates the approval result, updates the record, and invokes email notification actions.

Data Layer

ServiceNow records are used instead of an external database. REQ/RITM records preserve the service-request context, while the configured Database Tables table (u_database_tables) stores the operational network-request fields. Approval records are related through the configured Approval Request relationship.

4. Setup Instructions

Prerequisites

• Access to a ServiceNow Personal Developer Instance (PDI) or authorized ServiceNow instance.

• Permission to configure Service Catalog, Flow Designer, tables, roles and ACLs.

• Access to configure catalog variables, variable sets and Catalog UI Policies.

• Test users/roles for requester, approver and operational validation.

• Access to Flow Designer Execution Details for troubleshooting.

Configuration Sequence

1. Create the Network Request Service Catalog item under Networks and Connectivity.
2. Create and configure the catalog variables and the Requester Information variable set.
3. Configure the Catalog UI Policy for the relevant conditional field behaviour.
4. Create/configure the Database Tables (u_database_tables) table and required fields.
5. Configure the Create, Read, Write and Delete ACLs for the custom table.
6. Build the Network Request Flow Designer flow and configure Get Catalog Variables.
7. Map retrieved variables into the Database Tables operational record.
8. Configure Ask for Approval and the approved/rejected decision branches.
9. Configure request-created and approved/resolved email actions.
10. Configure the Approval Request relationship and validate REQ/RITM lifecycle tracking.
11. Run functional, performance, security and UAT validation.

5. Configuration / Component Structure

The supplied template contains a source-code folder structure for MERN. This ServiceNow project is configuration-driven, so the equivalent structure is shown below.

Configuration Area	Contents
Service Catalog	Network Request catalog item, Networks and Connectivity category, catalog variables and Requester Information variable set.
Client Configuration	Catalog UI Policy and catalog form behaviour.
Automation	Network Request Flow Designer flow, actions, conditions, approval and notification steps.
Data Model	REQ/RITM records and configured Database Tables (u_database_tables) operational table.
Approval	Ask for Approval action and Approval Request relationship to approval records.
Notifications	Send Email actions for request creation and approved/resolved handling.
Security	Roles and Create/Write/Read/Delete ACL records.
Operations / Diagnostics	Flow Designer Execution Details, platform logs and testing records.

6. Running the Application

There are no npm or local server commands because the application runs inside ServiceNow. To use the solution, open the configured ServiceNow Service Portal, navigate to the Network Request catalog item, enter the request information, submit the request, and follow the generated REQ/RITM and approval flow.

• Requester path: Service Portal → Service Catalog → Networks and Connectivity → Network Request → enter details → submit → track REQ/RITM.

• Administrator / configuration path: ServiceNow backend → Catalog / Flow Designer / tables / security → monitor configurations and executions.

• Approver path: authorized approval view → review the request → approve or reject according to configured access and process.

• Operational path: authorized record view → review the configured operational record and fulfilment information.

7. API Documentation

No external REST API endpoint is documented as part of the current implemented scope. The solution primarily uses native ServiceNow platform components and record interactions. Future integrations may expose or consume APIs for network-device automation, external messaging, analytics, or other enterprise systems.

Integration	Current Status	Purpose / Boundary	Future Interface
Network-device / infrastructure API	Future	Automated execution of approved network changes.	Network/device management API
Payment gateway	Not applicable to current network-request scope	No payment integration is implemented.	External payment API only if business scope expands
SMS / WhatsApp	Future	External request/status messaging.	Provider / WhatsApp Business API
Advanced analytics	Future	Reporting and operational analytics.	Analytics/BI platform

8. Authentication & Authorization

Authentication is provided by the ServiceNow instance. Authorization is implemented through roles and Access Control Lists (ACLs). Requesters use the Service Catalog experience, while authorized approvers, operational users and administrators access records according to configured permissions.

- Requester: submit a Network Request and view permitted request/RITM information.
- Approver: access approval information required for the configured approval process.
- Operational / fulfilment user: access the network-request operational record required for fulfilment.
- Administrator: configure catalog, flow, data model, notifications, roles and ACLs.
- ACL rules: restrict create, read, write and delete access on the configured Database Tables table according to role and record context.

9. User Interface

The primary user interface is the ServiceNow Service Portal catalog experience. The Network Request form captures requester and service details, applies the configured dynamic field behaviour, submits the request, and provides request tracking through REQ/RITM records. The current implementation uses the following visible request fields: Opened on behalf of, User Name, Email ID, Phone Number, Requested For, Mobile Number, Type of Connection, Existing ID, Total Amount, Mode of Payment, Address, Database Number, and Request Number.

10. Testing

Test Area	Coverage	Status
Catalog form	Network Request form, variables, required data and UI Policy behaviour	Pass
Request / RITM	Request submission, REQ/RITM creation and lifecycle tracking	Pass
Flow Designer	Catalog trigger, variable retrieval, record creation, approval, branching and execution path	Pass
Data mapping	Catalog variables to Database Tables operational fields	Pass
Approval	Ask for Approval, approval status and approved/rejected branching	Pass
Notifications	Request-created and approved/resolved email actions	Pass
Security	Roles and Create/Write/Read/Delete ACLs	Pass
Approval relationship	Operational record to approval-record relationship	Pass
UAT	End-to-end requester-to-approval-to-fulfilment journey	Pass

11. Screenshots or Demo

A final demonstration should show: Service Portal → Networks and Connectivity → Network Request → enter request details → submit → view REQ/RITM → review approval → inspect the configured Database Tables record → review notification/flow execution → follow the request lifecycle.

Project demo link: Not supplied in the documentation set. Insert the final demo video/deployed link before submission if required.

12. Known Issues

Issue / Limitation	Status	Resolution / Note
Approval configuration / approver visibility	Resolved	Validated Ask for Approval configuration and Approval Request relationship.
Variable-to-field mapping	Resolved	Corrected/verified catalog-variable retrieval and operational-field mappings.
Flow execution path validation	Resolved	Used Flow Designer Execution Details to diagnose and verify corrected paths.
UI visibility behaviour	Resolved	Validated Catalog UI Policy behaviour.
Production load benchmark	Not measured	No exact production-scale performance benchmark was supplied.
External network-device/API automation	Not implemented	Future scope.
Advanced analytics / BI	Not implemented	Future scope.
External messaging channels	Not implemented	Future scope.

13. Future Enhancements

- Automated integration with network-device or infrastructure management APIs.

• Automated fulfilment task creation and closure integration with network operations.
• Multi-level approval chains and policy-driven routing.
• SLA, escalation and advanced operational dashboards.
• External email/SMS/WhatsApp notification channels where required.
• AI-assisted request classification, priority recommendation or anomaly detection.
• Advanced network analytics and reporting.
• Expanded catalog coverage for additional network services.

Appendix - Functional Data Mapping

Data Element	Source	Destination	Purpose
Requested For	Catalog variable	Database Tables → Requested For	Identifies the employee/requester for whom the service is requested.
Mobile Number	Catalog variable	Database Tables → Mobile Number	Stores requester contact information.
Type of Connection	Catalog variable	Database Tables → Type of Connection	Captures whether the requested connection is New or Existing.
Existing ID	Catalog variable	Conditional form field	Captures existing connection/reference information when applicable.
Total Amount	Catalog variable	Database Tables → Total Amount	Stores the configured request amount.
Mode of Payment	Catalog variable	Database Tables → Mode of Payment	Stores the selected payment mode value captured by the form.
Address	Catalog variable	Database Tables → Customer Address	Stores the requested service/customer address.
Database Number	Catalog variable	Database Tables → Database Number	Stores the configured database/reference number.
Request Number	Catalog variable / request context	Database Tables → Request Number	Supports request identification and tracking.
Approval Status	Flow / approval result	Database Tables → Approval Status	Tracks Requested, Approved or Rejected state.
REQ/RITM Reference	ServiceNow request records	Request / Requested Item	Preserves service-request context and lifecycle tracking.